Code No: 111AL **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year Examinations, March/April - 2023
MATHEMATICAL METHODS
(Common to EEE, ECE, CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a) Show that $E - 1 = (1 - \nabla)^{-1} - 1$. [2]

b) If $f(x) = \sin(px + q)$, find $\Delta^2 f(x)$. [3]

c) If the first two approximations for the root of $x^3 - x^2 + 1 = 0$ are -1 and 0, find the next two approximations using Regula falsi method. [2]

d) The values of x and f(x) are given below. Find $\int_0^{\pi/2} f(x)dx$. [3]

x	0	$\pi/6$	$\pi/3$	$\pi/2$
f	1	1.6487	2.3774	2.71828

e) If *f(x)= |sinx| in (-L,L),* then find the Fourier coefficient b_n. [2]

f) Find the finite Fourier cosine transform of $f(x) = x^2$, 0<x< 4. [3]

g) Find the general solution of $pqz = p^2(qx + p^2) + q^2(py + q^2)$. [2]

h) Solve *px+qy=1* using the method of separation of variables. [3]

i) If $\phi = ax^2 + by^2 + cz^2$ satisfies Laplacian equation, then find the value of a + b + c. [2]

j) If $\bar{F} = ax\bar{i} + by\bar{j} + cz\bar{k}$ where a, b, c are constants evaluate $\iint \bar{F}.\bar{n}ds$, where S is the surface of the unit sphere. [3]

PART - B

(50 Marks)

2.a) Find f(2.4) from the following data using appropriate interpolation method.

x	1.0	1.5	2.0	2.5
f(x)	3	3.375	5.0	12.072

b) Fit a second degree polynomial to the following data by the method of least squares.[5+5]

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

OR

3.a) Using Lagrange's interpolation formula find y(8) from the following table:

X	5	6	9	11
Y	12	13	15	18

b) Fit a $y = ab^x$ by the method of least squares to the data given below. [5+5]

x	0	1	2	3	4	5	6	7
y	10	21	35	59	92	200	400	610

4.a) Solve the system *10x-2y-z-w=3; -2x+10y-z-w=15; -x-y+10z-2w=15; -x-y-2z+10w= -9* using Gauss Seidel method.

b) Find the root of the equation $x\log_{10}(x)=1.2$ using False position method. [5+5]

OR

5.a) Evaluate $\int_0^2 e^{-x^2}dx$ using Simpson's rule taking h = 0.25.

b) Using Milne's method, find *y(4.5),* given $5xy' + y^2 - 2 = 0$, *y(4)=1, y(4.1)= 1.0049, y(4.2)= 1.0097, y(4.3) =1.0143 and y(4.4)= 1.0187.* [5+5]

6.a) Find the Fourier series of $f(x)=\left(\frac{\pi-x}{2}\right)^2$ in the interval $0<x<2\pi$.

b) Find the finite Fourier sine transform of $f(x)=\frac{x}{\pi}, 0<x<\pi.$ [5+5]

OR

7.a) Obtain the Fourier cosine series for f(x) = x sin x, $0<x<\pi$

b) Find the Fourier transform of $f(x)=\frac{1}{\sqrt{|x|}}$. [5+5]

8.a) Find the partial differential equation arising from $\phi(x+y+z, x^2+y^2+z^2)=0$.

b) Solve $px(z-2y^2)=(z-qy)(z-y^2-2x^3)$. [5+5]

OR

9.a) Solve $\frac{\partial z}{\partial x}-4\frac{\partial z}{\partial y}=0,\ z(0,y)=8e^{-3y}$ by the method of separation of variables.

b) Solve $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ subject to the conditions u(0, y) = 0, u(l, y) = 0, u(x, 0) = 0 and u(x, a) = $\sin\frac{n\pi x}{l}$, where $0\le x\le l, 0\le y\le a$ and n is a positive integer. [5+5]

10.a) Find the angle of intersection of the spheres $x^2 + y^2 + z^2 = 39$ and $x^2 + y^2 + z^2 + 4x - 6y - 8z + 52 = 0$ at the point (4, -3, 2).

b) Evaluate $\int_C \bar{F}.d\bar{r}$ where $\bar{F}=3xy\bar{i}-y^2\bar{j}$ and C is the curve $y = 2x^2$ in xy-plane from (0, 0) to (1, 2). [5+5]

OR

11.a) Find the directional derivative of xyz^2+xz at (1, 1, 1) in a direction of the normal to the surface $3xy^2+y=z$ at (0, 1, 1).

b) Evaluate $\iint_S \bar{F}.\bar{n}\,ds$, if $\bar{F}=yz\bar{i}+2y^2\bar{j}+xz^2\bar{k}$ and S is the surface of the cylinder $x^2 + y^2 = 9$ contained in the first octant between the planes z = 0 and z = 2. [5+5]

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